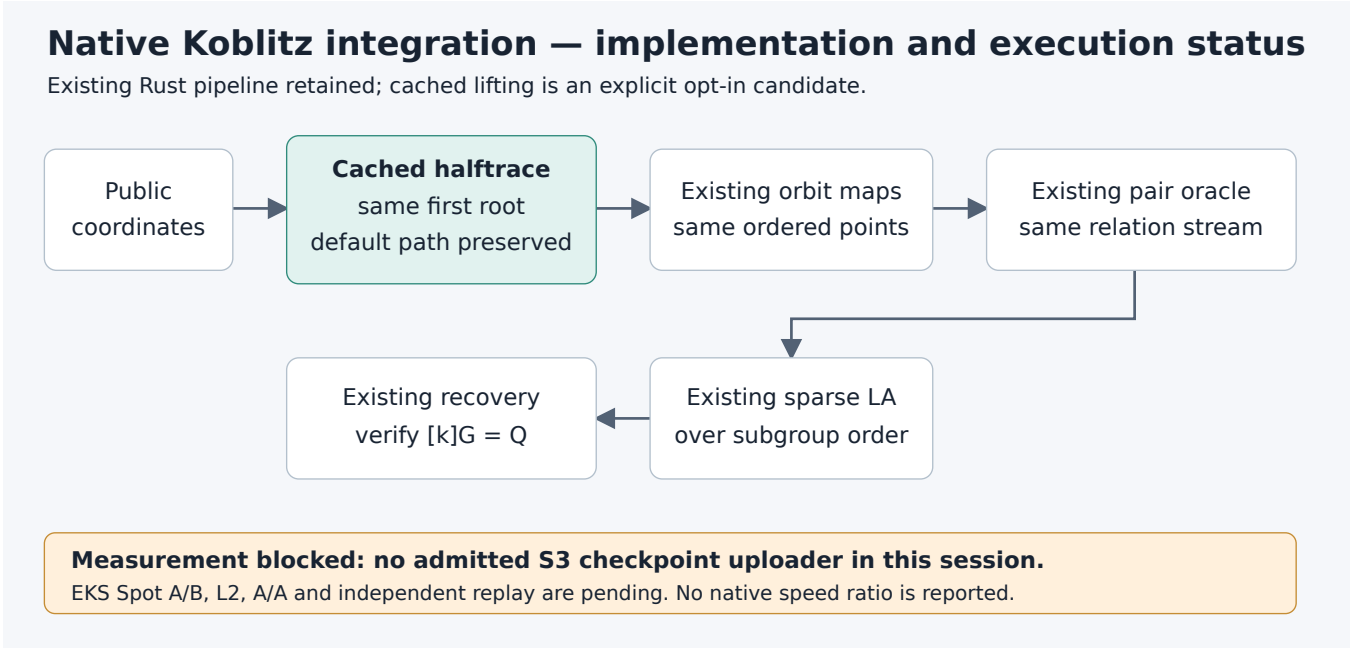
Native Koblitz cached lifting

2026-10-30 · Engineering implementation candidate · Generic decomposition track

Measurement pending. The native baseline was built before source edits. The mandatory S3 checkpoint upload failed because this session has no approved uploader. No scientific EKS job was launched. No native speed ratio is claimed.



Implementation

Lazy, curve-local byte tables evaluate the existing odd-degree halftrace. The explicit factor-base parameter `halftrace_tables=1` preserves the first root, ordered points, signed orbit maps and existing relation/LA implementation. Default recipes retain the original point-lifting path. Initialized immutable tables are shared by clones. No field-wide table or new Python path is introduced.

Verification and accounting

Native checks compare exhaustive small-field roots and sampled wider roots to the generic field definition, test scope restoration and field separation, and compare ordered point lifts including non-Koblitz coefficients. An end-to-end fixture compares two- and three-summand known-answer solves; a second test exercises the public ecbench pipeline and its point hash and counters. Both tests pass, and strict library Clippy passes. Full-suite completion is tracked separately under `development/`; development elapsed times are not benchmarks.

The frozen comparison has eight independent one-target workloads, the same base and seed in both arms, strong-rho and IC A/A controls, six alternating measured rounds and one warmup. EKS Spot L2 admission and independent scalar replay are mandatory. Cache construction belongs in the cold setup cost.

Limits and sources

The degree-31 invariant subspace has ord_31(2)=5; it does not discharge the degree-83 confidence gate or establish transfer to ECC2K-130. Existing coarse Artin–Schreier solve counters alone do not price the cache delta. External arithmetic comparisons have not been measured. The scoreboard and leaderboard remain unchanged because no admitted performance result exists.

Baseline source: `3b34d504bc13ce668e86d8e68cdf57e3fbaba89a`. The primary binary point-recovery contract is SEC 1 v2.0, section 2.3.4. Existing `src/binary_ecc/f2m.rs` supplies the native trace/halftrace precedent. Implementation paths: `src/cryptanalysis/koblitz_fast.rs`, `src/cryptanalysis/koblitz_index_calculus.rs`, and `src/cryptanalysis/ic_framework/plugins.rs`. The study's `README.md`, `PROTOCOL.md`, `spec.json` and `BASELINE.txt` retain the detailed contract and evidence links. The exact access gap is recorded in `problems/native-koblitz-measurement-access.md`.